SECTION 1 — Quantitative investment baseline

Uploaded-document evidence: The uploaded 2025 Annual Report provides a highly detailed financial and operational baseline that contextualizes the company's overall expenditures, capital investments, software capitalization, and restructuring activities across a multi-year period. For the fiscal year ended December 31, 2025, the company reported total net sales of $14,316 million, representing a 2% year-over-year increase from $14,086 million in 2024, which in turn was slightly down from $14,198 million in 2023.[1] The company explicitly categorizes its revenue generation into two primary technology and product segments: eProducts, which encompass all products utilized on or for electric vehicles and adaptable hybrid powertrains, and Foundational products, which encompass traditional internal combustion engine components and hybrid applications.[1] In 2025, the eProducts portfolio generated $2,570 million, constituting 18% of total revenue, which represents an increase from $2,335 million (17%) in 2024 and $2,037 million (14%) in 2023.[1] Foundational products accounted for $11,746 million, or 82% of total revenue in 2025, indicating a slight decline from $11,751 million (83%) in 2024 and $12,161 million (86%) in 2023.[1]

Investment intensity across the enterprise is captured through several distinct financial line items, beginning with research and development (R&D) and engineering expenses. Gross R&D expenditures were reported at $823 million in 2025, marking a decrease from $872 million in 2024 and $856 million in 2023.[1] The company offsets a portion of these gross engineering costs with direct customer reimbursements, which totaled $113 million in 2025, $136 million in 2024, and $139 million in 2023.[1] Consequently, net R&D expenditures were $710 million in 2025, down from $736 million in 2024 and $717 million in 2023.[1] As a percentage of net sales, net R&D expenditures stood at 5.0% in 2025, reflecting a slight contraction from 5.2% in 2024 and 5.1% in 2023.[1] The filing explicitly notes that this decrease in net R&D costs was primarily due to a decreasing net investment related to the company's eProducts portfolio.[1]

Capital expenditures (capex), which include tooling outlays and physical infrastructure investments, were reported at $469 million for the year ended December 31, 2025.[1] This represents a significant year-over-year decrease from $671 million in 2024 and $832 million in 2023.[1] Capital expenditures as a percentage of sales dropped from 4.8% in 2024 to 3.3% in 2025.[1] The financial filing explicitly states that the year-over-year decline in capital expenditures primarily reflects lower eProduct investments, aligning with broader market volatility and delays regarding global electric vehicle adoption.[1]

The company also reports significant restructuring and impairment charges that contextualize its operational technology investments and strategic realignments. In 2025, the company recorded $101 million in restructuring expenses, an increase from $74 million in 2024 and $79 million in 2023.[1] This 2025 restructuring figure includes $31 million tied specifically to the 2024 Structural Costs Plan, which was designed to address cost structures in the PowerDrive Systems segment due to electric vehicle market volatility, aiming for approximately $100 million in annual cost savings by 2026.[1] Furthermore, the company recorded massive impairment charges totaling $624 million in 2025, following $646 million in impairments in 2024 and $29 million in 2023.[1] The 2025 impairments included a $423 million goodwill impairment related to the Battery & Charging Systems reporting unit, driven by dual sourcing actions by customers, pricing pressures from competitors, and EV adoption delays in North America.[1] The 2024 impairments included $577 million related to goodwill at both the PowerDrive Systems and Battery & Charging Systems segments.[1]

A major strategic portfolio decision directly impacting digital and connected product investments was made in February 2025, when the company decided to completely exit its charging business within the Battery & Charging Systems reportable segment.[1] Production operations ceased during the second quarter of 2025, generating $32 million in total charges.[1] These charges included a $22 million loss on the sale of the Hubei Surpass Sun Electric (SSE) business (which was sold for $7 million in proceeds), the write-off of $9 million of inventory, and impairments of intangible assets ($22 million), goodwill ($13 million), and fixed assets ($4 million).[1]

Regarding capitalized software and intangible assets, the annual report notes that amortization of other intangible assets, which includes patented and unpatented technology as well as customer relationships, was $66 million in 2025, down slightly from $69 million in 2024 and $67 million in 2023.[1] The total net carrying amount of amortized intangible assets stood at $390 million at the end of 2025, compared to $471 million at the end of 2024.[1] While the company does not disaggregate internal-use software capitalization directly in the primary expense tables, it notes that pre-production costs for long-term supply arrangements and tooling are capitalized within property, plant, and equipment, and that the company evaluates the impact of Financial Accounting Standards Board (FASB) Accounting Standards Update (ASU) 2025-06 regarding targeted improvements to the accounting for internal-use software.[1]

The uploaded hiring dataset quantifies talent acquisition signals related to artificial intelligence, machine learning, agentic systems, and adjacent digital capabilities over a multi-year period. The dataset captures 6,932 total observed job postings.[1] Within this population, 20 postings are classified as AI-core, meaning AI, ML, LLMs, GenAI, or MLOps are explicit and central to the role, and 79 are classified as AI-adjacent.[1] The hiring data translated into prose reveals a slow but steady relative increase in AI-core roles as a percentage of total hiring volume over time. In 2022, 0 out of 456 total postings (0.0%) were AI-core.[1] In 2023, AI-core postings rose to 3 out of 1,073 (0.3%).[1] In 2024, the volume reached 4 out of 1,291 postings (0.3%).[1] In 2025, AI-core postings expanded to 7 out of 2,723 (0.3%).[1] By the partial year of 2026, AI-core postings accounted for 6 out of 1,365 postings, reaching a peak concentration of 0.4% of total hiring volume.[1]

The capability mix within these AI-core postings demonstrates a highly targeted technical focus. In 2023, all 3 AI-core roles were dedicated exclusively to AI/ML modeling.[1] In 2024, all 4 AI-core roles were dedicated to AI/ML modeling.[1] In 2025, all 7 roles remained focused on AI/ML modeling.[1] However, in 2026, the dataset shows a diversification in capability focus: 4 roles were dedicated to AI/ML modeling, and 2 roles emerged specifically for Agentic AI.[1] No explicitly

AI-core roles were observed in the dataset for industrial automation, supply chain optimization, or corporate functions.[1]

Public-source evidence: Trade publications, vendor documentation, and industry case studies provide additional context regarding the company's software, cloud, and digital infrastructure expenditures. The company's digital transformation costs and IT budgets are partially managed through targeted vendor platforms. For example, IT financial management and budgeting are handled through Nicus Software deployed within a ServiceNow environment.[4] The company utilized this software to consolidate IT spending forecasts previously managed by over 20 IT managers across separate spreadsheets, a process that historically expanded the budget season into a ten-month cycle.[4] Public records also indicate that the company maintains core enterprise resource planning software via SAP, specifically undertaking a migration framework for data exchange between Siemens Teamcenter and SAP S/4HANA to automate the exchange of change objects, parts, documents, and manufacturing bills of materials.[6]

The company engages in cloud computing and data platform expenditures through partnerships with major cloud providers and specialized storage vendors. The company transitioned 100 global locations to hybrid cloud storage using the Nasuni File Data Platform, a move designed to centralize petabytes of file data, accelerate global engineering workflows, and provide data protection and recovery mechanisms.[7] Furthermore, the company utilizes Amazon Web Services (AWS) and Microsoft Azure environments to host generative AI workloads, digital product delivery pipelines, and engineering data.[6]

Assessment:

The company's financial disclosures reveal a traditional, hardware-centric automotive supplier undergoing a highly scrutinized and financially challenging transition toward electrification and software-defined architectures. The financial evidence does not isolate AI, cloud, or enterprise software as standalone reporting segments, embedding these technological costs within broader R&D, SG&A, and capex figures. The significant reductions in both net R&D and capital expenditures in 2025, coupled with over $1.2 billion in cumulative impairments across 2024 and 2025, indicate a highly constrained investment environment driven by electric vehicle market volatility. The strategic exit from the charging business, which resulted in the write-off of associated intangible assets and inventory, further reflects strict capital discipline and a narrowing of the product portfolio.

Despite these severe financial contractions and restructuring efforts, the uploaded hiring dataset indicates that investments in specialized AI talent have remained resilient, slowly increasing as a percentage of total hiring from 0.0% in 2022 to 0.4% in 2026. The absolute volume of AI roles (20 AI-core postings over a five-year span) is objectively small relative to the total enterprise workforce of 37,500 employees. This disparity between the massive overall corporate headcount and the highly concentrated AI hiring signals suggests that AI capability is highly centralized. Rather than representing a massive, company-wide organizational shift, the evidence supports the conclusion that the company is exercising strict capital discipline, utilizing targeted AI, machine learning, and enterprise software investments to incrementally optimize existing engineering, simulation, and manufacturing processes.

SECTION 2 — Capability-by-capability analysis, including GenAI and agentic systems

Machine Learning for Engineering Simulation and Model Order Reduction

Uploaded-document evidence: The uploaded hiring dataset indicates a profound concentration of AI investments directed toward engineering and simulation optimization. The company is actively funding machine learning models and deep learning capabilities designed to accelerate and optimize complex product design.[1] Specific technical applications cited in the dataset include the utilization of neural network structures to perform "model order reduction of electro-thermal physical models," predict outcomes for e-motor simulations, and execute aerodynamic simulations for turbomachinery.[1] The dataset explicitly states that the engineering problem being solved is the need to predict critical physical positions and performance metrics using Finite Element Analysis (FEA) and Computer-Aided Design (CAD) data without requiring computationally expensive and time-consuming full simulations.[1] This capability aims to fundamentally improve the speed and efficiency of product analysis. Furthermore, the dataset reveals that out of 20 total AI-core postings, 13 are dedicated to a "Materials Research" business area, representing the largest single pocket of AI talent acquisition within the company.[1]

Public-source evidence: Public engineering documentation, published research papers, and academic partnerships heavily corroborate the company's deployment of machine learning for engineering simulation and modeling. The company utilizes high-fidelity multiphysics numerical modeling platforms, such as Amesim and COMSOL, to generate baseline reference datasets.[10] These massive datasets, which are verified using real hardware tested in physical vehicle environments, are then used to train artificial intelligence and machine learning models to act as "Deep learning virtual sensors".[10] This technique, formally known as Reduced Order Modeling (ROM), is applied to evaluate the impact of highly specific variables—such as variable switching frequencies and pulse width modulation strategies—on the efficiency, cost, and energy consumption of subcomponents within the company's next-generation electric drive systems.[10] Additional public evidence demonstrates the company's application of machine learning to the physical control of motors and clutches. In partnership with academic researchers at its Landskrona, Sweden facility, the company has actively explored replacing traditional Proportional-Integral (PI) controllers and Pulse Width Modulation (PWM) techniques with neural networks to better compensate for nonlinearities in Brushless DC (BLDC) permanent magnet motors utilized in its actuators.[11] Similarly, the company has utilized machine learning and system identification techniques to estimate temperatures within wet lamella clutches.[12] Because not all temperatures can be measured physically during operation, the company

updates existing thermal models using MATLAB Simulink to accurately estimate temperatures in newer products without requiring physical temperature sensors in all locations.[12] Furthermore, published research funded by the company details the use of Proper Orthogonal Decomposition (POD) and Galerkin projection-based model order reduction to simulate evaporator heat exchangers in organic Rankine cycle waste heat recovery systems, allowing engineers to drastically reduce computation time while maintaining accuracy.[14]
Assessment:
The evidence supports capability building transitioning into pilot activity and early internal operational deployment. The company is actively developing internal capabilities to use neural networks, deep learning, and advanced mathematical projections to bypass the computational bottlenecks of traditional FEA and thermal simulations. By creating Reduced Order Models and virtual sensors, the company solves the critical engineering problem of lengthy simulation cycles, allowing R&D teams to rapidly iterate on e-motor, turbomachinery, and power electronics designs without building physical prototypes for every iteration. The evidence indicates these are internally developed capabilities that heavily utilize vendor-provided baseline software tools (Amesim, COMSOL, MATLAB/Simulink) to generate the necessary training data and validate the machine learning outputs.

Generative AI and Agentic Systems for Engineering Workflows

Uploaded-document evidence: The uploaded hiring dataset reveals the very recent emergence of Generative AI and autonomous agentic systems as a distinct capability pocket. In 2026, the dataset recorded 2 AI-core roles specifically dedicated to "Agentic AI," the first appearance of this category in the multi-year dataset.[1] The dataset narrative indicates that the company is investing in "agentic AI systems and AI-driven tools to meet 2026 engineering efficiency targets".[1] These systems are intended to support technology scouting, manage the end-to-end deployment of AI projects, and translate complex business problems into actionable technical solutions.[1] Furthermore, the dataset notes that the company is exploring the deployment of "AI agent and multi-agent systems within real products and engineering workflows" to enhance mobility solutions and inform the strategic direction of the product portfolio.[1]
Public-source evidence: Public recruitment data, architectural documentation, and partner case studies heavily corroborate the active development and deployment of agentic systems, specifically centered at the company's facility in Darmstadt, Germany. The company is actively hiring Senior Software Development Engineers to build "production-grade AI agent and multi-agent software capabilities".[15] The engineering problem being solved is the safe and reliable integration of non-deterministic AI components with highly deterministic embedded and enterprise systems.[15] The evidence shows that these proprietary systems utilize Large Language Models (LLMs), advanced prompt engineering, data embeddings, and tool/function calling mechanisms.[15] To ensure the reliability and safety of these multi-agent workflows in a strict engineering environment, the company is deploying rigorous evaluation frameworks, including offline evaluations, regression suites, red-teaming, and observability tracing for

debugging multi-step tool workflows.[16]
Additionally, vendor case studies indicate that the company has utilized third-party expertise to support broader Generative AI implementation and IT support. A partner case study with Arcanum AI details how the company securely migrated generative AI workloads from OpenAI to AWS, utilizing Amazon Bedrock to establish a private network for handling sensitive client data and deploying open-source LLMs evaluated by internal experts.[8] For basic IT service management, the company utilizes an IT Virtual Agent chatbot named "Indy," which is hosted within the ServiceNow (NOW) system to autonomously handle end-user help requests, process IT catalog items, and monitor customer experience dashboards based on user feedback.[18]
Assessment:
The evidence supports pilot activity and active capability building for advanced agentic engineering workflows, alongside vendor-supported operational deployment for basic GenAI IT support. The company's hiring of Senior Software Development Engineers for AI Agents in Darmstadt demonstrates a serious, internally funded commitment to building proprietary multi-agent systems that interact directly with core engineering platforms. The explicit technical focus on red-teaming, tool calling, and non-deterministic integration indicates that the company clearly recognizes the risks of LLM hallucination in safety-critical engineering contexts and is actively engineering software guardrails to mitigate these risks. The migration of workloads to AWS Bedrock confirms that the company is prioritizing data security and private cloud infrastructure for its enterprise LLM deployments, while the use of the "Indy" chatbot demonstrates a pragmatic use of vendor GenAI for internal IT cost reduction.

Software Engineering Platforms and Embedded Systems (AUTOSAR/ASPICE)

Uploaded-document evidence: The uploaded hiring dataset indicates that the company is building data analysis and statistical methods to optimize product characteristics using measurement data.[1] The evidence notes the development of machine learning and data analytics algorithms specifically for propulsion management systems, designed to optimize the performance of connected and automated vehicles.[1]
Public-source evidence: Public evidence reveals a highly mature, scaled internal deployment of embedded software engineering practices, governed by strict automotive industry standards. The company relies heavily on the AUTOSAR (AUTomotive Open System ARchitecture) framework and the Automotive SPICE (ASPICE) process maturity model for the development of real-time embedded systems.[20] The operational problem solved by these capabilities is the fundamental need for safe, reliable, and standardized communication between electronic control units (ECUs), inverters, converters, and actuators within complex power electronics and drivetrain systems.[22]
The company employs dedicated software technical leaders to manage the architecture, design, and development of Basic Software (BSW), Low-Level Drivers (LLD), Complex Device Drivers (CDD), and hardware abstraction layers interfacing with sophisticated automotive microcontrollers, specifically citing the Infineon Aurix Tricore series.[22] Engineering workflows utilize model-based design methodologies, relying on tools such as MATLAB, Simulink,

TargetLink, and Stateflow to conduct Model-in-the-Loop (MIL) and Software-in-the-Loop (SIL) unit testing and auto-coding.[22] Furthermore, the company utilizes industry-standard vendor configuration tools such as Vector DaVinci Configurator, DaVinci Developer, and Elektrobit (EB) tresos.[22] To manage software requirements, source code, and product lifecycle data globally across multi-site agile teams, the company utilizes Siemens Polarion ALM and Siemens Teamcenter, while undertaking large-scale data migration efforts to move certain product lifecycle data from Siemens Teamcenter to PTC Windchill.[25]

Assessment:

The evidence supports scaled, production deployment of embedded software capabilities. Unlike the exploratory agentic AI systems, the development of AUTOSAR-compliant basic software and application software represents the core, highly mature digital nervous system of the company's physical products. The evidence demonstrates that the company maintains deep internal ownership of these capabilities, utilizing a standardized vendor toolchain (Simulink, Vector DaVinci, EB tresos) to write, test, and integrate embedded C code for safety-critical propulsion hardware. The hiring of specialized technical leaders to manage globally distributed agile teams confirms that embedded software engineering is a fully scaled operational function critical to the company's commercial offerings.

Manufacturing AI, Machine Vision, and Industrial IoT

Uploaded-document evidence: The uploaded hiring dataset does not show specific evidence of manufacturing AI, computer vision, or factory-floor automation roles among the AI-core postings.[1]

Public-source evidence: Public evidence, primarily from technology vendors and industrial partners, demonstrates that the company is actively deploying AI and machine vision to solve manufacturing quality and production efficiency problems. At the company's production center in Changnyeong, South Korea, electric vehicle power drive units are assembled and inspected for defects.[28] To ensure absolute precision on the assembly line, the company deployed Cognex 3D and 2D machine vision systems, specifically utilizing the In-Sight 2800 system.[28] This system utilizes AI-powered inspection algorithms to detect missing parts on highly flat surfaces and verify the correct physical assembly of the Park-Lock mechanism—a critical mechanical safety component that prevents unintended vehicle movement when parked.[28] The use of computer vision solves the operational problem of manual inspection bottlenecks and human error in highly complex EV component assembly.[29]

Furthermore, the company utilizes Industrial Internet of Things (IIoT) platforms to monitor energy consumption across its global manufacturing footprint. Partnering with Siemens, the company deployed the Siemens Insights Hub (formerly MindSphere) along with a specialized energy management application.[30] This cloud-based IIoT platform collects real-time operational data from 63 manufacturing locations across 20 countries, successfully replacing historically manual data collection efforts.[31] This deployment addresses the strategic problem of baselining, measuring, and reducing global energy use to meet corporate carbon neutrality targets, providing plant managers with detailed analytics dashboards.[30]

Assessment:

The evidence supports vendor-supported operational deployment of machine vision and Industrial IoT capabilities. The company is not building proprietary computer vision foundation models internally; rather, it is buying and integrating mature, off-the-shelf industrial solutions from specialized vendors like Cognex and Siemens. These systems are operationally deployed in live production environments (such as the Changnyeong facility) and scaled across the global manufacturing footprint (as seen with the 63-site Siemens deployment). This indicates a highly pragmatic approach to factory-floor digital transformation, leveraging partner technology to achieve immediate return on investment in quality assurance, defect reduction, and sustainability tracking.

Enterprise Data Analytics and IT Financial Auditing

Uploaded-document evidence: The uploaded hiring dataset indicates that the company is developing an enterprise data analytics and AI/ML platform.[1] This platform is intended to provide advanced analytics capabilities to internal data teams across the organization, supporting operational efficiency, translating business problems into technical solutions, and aiding strategic decision-making.[1]

Public-source evidence: Public evidence confirms the scaled deployment of specialized analytics platforms to optimize enterprise resource planning, financial auditing, and IT financial management. The company utilizes a vendor analytics platform called Supervizor to bypass the slow, resource-intensive process of building financial data analytics internally.[32] Supervizor is integrated directly into the company's SAP environment across 20 company codes globally.[32] It utilizes AI-powered analytics and over 350 pre-built routines to automate Procure-to-Pay (P2P) testing, detecting duplicate invoices and other key risk indicators.[32] This solves the commercial problem of manual audit sampling, allowing the internal audit team to filter out thousands of low-risk alerts and focus exclusively on high-risk root-cause investigations, which the evidence claims resulted in direct cash-recovery opportunities within the first six months of deployment.[32]

For IT financial management, the company deployed software from Nicus to fundamentally transform its internal budgeting processes.[4] Previously, IT annual planning required approximately six months and relied on disparate spreadsheets managed by over 20 IT managers, a process that historically expanded the budget season into a chaotic ten-month cycle.[4] By integrating Nicus within a ServiceNow environment, the company automated its Bill of IT and cost transparency models, mapping technology costs directly to business goals and eliminating spreadsheet version control issues.[5] To support global data availability and infrastructure modernization, the company transitioned 100 locations to hybrid cloud storage using the Nasuni File Data Platform, centralizing petabytes of file data to accelerate global engineering workflows and protect against ransomware attacks.[7]

Assessment:

The evidence supports scaled, vendor-supported operational deployment of enterprise analytics and data infrastructure. The company is actively optimizing its back-office operations—specifically financial auditing, IT budgeting, and global file storage—by integrating specialized Software-as-a-Service (SaaS) and AI-driven platforms. The evidence demonstrates measurable business impact, including reduced audit times, streamlined budget consolidation, and improved data recovery capabilities. The company relies entirely on external vendors (Supervizor, Nicus, Nasuni, ServiceNow) to provide the core technology, focusing its internal IT efforts on systems integration, process adoption, and vendor management.

Connected Services and EV Charging Cloud (Exited Capability)

Uploaded-document evidence: The uploaded hiring dataset and the 2025 Annual Report do not show active investment in connected EV charging services. The 2025 Annual Report explicitly states that in February 2025, the company made a strategic portfolio decision to exit its charging business, completely ceasing production operations in the second quarter of 2025.[1]

Public-source evidence: Prior to the 2025 exit, public vendor evidence indicated that the company had collaborated extensively with Deloitte to explore a cloud-based, connected EV charger solution.[34] This initiative aimed to shift the company from supplying under-the-hood components to managing software-enabled products that interacted directly with end consumers.[34] Deloitte engineered a software technology stack utilizing its IndustryAdvantage digital assets, built a custom data model hosted on AWS, and created dashboards to track charging sessions and identify fault codes for proactive maintenance.[34]

Assessment:

The evidence supports that this was an exploration and pilot activity that has since been strategically divested. While the Deloitte partnership successfully built a proof-of-concept cloud architecture for connected chargers, the company's financial filings confirm the complete cessation of the charging business in 2025, including the sale of the SSE unit and the write-off of associated intangible assets. Therefore, this capability is no longer an active area of internal investment, deployment, or strategic focus.

SECTION 3 — Talent, teams, and operating model

Uploaded-document evidence: The uploaded hiring dataset, comprising 6,932 total observed postings, provides a detailed geographic, organizational, and seniority map of the company's AI and digital talent acquisition efforts. The geographic footprint is highly distributed across multiple continents, though the absolute volume of AI-specific roles remains objectively small. The United States accounts for the largest overall posting volume with 2,067 total postings.[1] Within the US, 15 roles were identified as AI-core, yielding an AI intensity of 0.73%.[1] Poland represents the second-largest overall hiring hub with 827 total postings, followed by Mexico (796) and Portugal (562).[1] Germany, with 517 total postings, hosted 4 AI-core roles, representing an AI intensity of 0.77%.[1] India (392 postings) and China (253 postings) also represent significant overall hiring bases, though the dataset does not isolate their AI-core denominators.[1]

Organizationally, the AI-core postings are heavily concentrated in R&D and product engineering functions. Out of the 20 AI-core postings, 13 are located within the "Materials Research" business area, representing an AI intensity of 1.14% against a denominator of 1,140 total postings in that function.[1] "Customer Offerings" accounted for 4 AI-core roles out of 691 (0.58%), and "IT and Data" accounted for 3 AI-core roles out of 293 (1.02%).[1] The dataset shows absolutely no AI-core postings in Supply Chain, Sales/Marketing, Manufacturing Operations, or Corporate Functions.[1]

The seniority mix of the AI-core roles skews heavily toward junior talent. Of the 20 AI-core postings, 14 were classified as Junior, 3 as Mid Level, and 3 as Senior.[1] In terms of capability alignment by business area, AI/ML Modeling roles were overwhelmingly located in Materials Research (12 roles), with a smaller presence in Customer Offerings (3) and IT and Data (3).[1] The emerging category of Agentic AI Systems (2 roles total in 2026) was split evenly, with one role in Customer Offerings and one role in Materials Research.[1]

Public-source evidence: Public evidence aligns with and expands upon the geographic and organizational signals found in the hiring data, providing concrete examples of what these teams are actually building. The presence of Senior Software Development Engineers focused on AI Agents in Darmstadt, Germany [16] directly correlates with the "Agentic AI" postings identified in the dataset and explains the German AI-core concentration. The public evidence also highlights Changnyeong, South Korea, as a critical operational hub for manufacturing engineering and production technology, specifically utilizing Cognex machine vision teams on the factory floor.[28] Auburn Hills, Michigan, serves as a corporate IT hub, evidenced by hiring for IT Service Desk roles supporting the ServiceNow platform and the "Indy" virtual agent.[18] Furthermore, engineering roles for embedded software (AUTOSAR) are highly globalized, with job postings indicating strict requirements for multi-site agile team collaboration and cross-border integration.[20]

The operating model relies on a distinct hybrid approach to capability building. For core product engineering—such as embedded Basic Software (BSW), neural networks for model order reduction, and proprietary AI agent architectures—the company relies almost entirely on internal engineering teams.[10] Conversely, for enterprise IT, cloud data infrastructure, and factory-floor automation, the company relies heavily on external vendors and consultants. Partnerships with Siemens (Insights Hub), Cognex (machine vision), Supervizor (financial auditing), Nicus (IT budgeting), and AWS (generative AI hosting) demonstrate a strategy of buying off-the-shelf digital capabilities rather than building them internally.[4]

Assessment:

The evidence suggests a highly targeted, engineering-centric operating model for AI and software talent. The company is not undertaking a mass hiring spree of AI generalists or data scientists across all corporate functions; rather, it is inserting specialized AI practitioners almost exclusively into R&D (Materials Research) and product development (Customer Offerings). The heavy concentration of junior AI roles indicates a strategic capability-building phase where the company is acquiring foundational data science skills to supplement its existing mechanical and electrical engineering base, rather than importing top-heavy AI leadership.

However, the recent appearance of Senior AI Agent roles in Germany suggests a maturation in specific capability pockets, moving from basic predictive modeling to complex, multi-agent software architecture. Geographically, the company leverages a global footprint, utilizing the US and Germany for advanced R&D and AI architecture, while leveraging regions like India, Poland, and Mexico for broader engineering and IT execution. The stark division between internal capability building for product software (AUTOSAR, e-motor ML) and vendor reliance for enterprise software (SAP, ServiceNow, Supervizor) reflects a disciplined operating model focused on protecting core intellectual property while outsourcing generic digital workflows to lower costs.

SECTION 4 — What the evidence implies for investment priorities and competitor monitoring

Based on the synthesis of financial disclosures, hiring signals, and public engineering documentation, the company's investments in AI, software, and digital capabilities reveal three distinct strategic priorities.

Priority 1: Accelerating engineering velocity through Model Order Reduction to offset capital constraints

The company is strategically prioritizing the use of artificial intelligence and machine learning to reduce the time, cost, and computational burden associated with product R&D, specifically through Model Order Reduction and neural network simulation.

Uploaded-document evidence: Financial filings reveal a highly constrained capital environment, marked by a deliberate reduction in both net R&D and capital expenditures in 2025 due to electric vehicle adoption volatility, alongside over $1.2 billion in cumulative impairments across eProduct segments.[1] Simultaneously, the hiring dataset shows a sustained, targeted investment in AI/ML modeling within Materials Research, specifically designed to bypass full FEA and CAD simulations using neural networks.[1]

Public-source evidence: The company uses high-fidelity platforms like Amesim and COMSOL to train deep learning virtual sensors, allowing engineers to rapidly test variables like switching frequencies in e-motors without building physical prototypes or running exhaustive 3D physical simulations.[10] The company is also building proprietary AI agent workflows in Darmstadt, Germany, to assist engineers in translating complex business problems into technical solutions.[16]

Assessment:

The company appears to be using AI not as a standalone commercial end-product, but as an internal margin-defense mechanism. By using deep learning and mathematical projection to compress simulation times, the company seeks to maintain its product development velocity and engineering output despite recent cuts to gross R&D and capex budgets. What is known is that the company has the foundational data (from Amesim/COMSOL) and the specific talent (AI Agent engineers in Darmstadt) to execute this strategy. What remains uncertain is whether these Reduced Order Models and virtual sensors are accurate enough to entirely replace physical testing in safety-critical power electronics and thermal management systems.

Implication for a competitor: Competitors (such as Bosch, Valeo, and Magna) must recognize that the company is attempting to decouple its engineering output from its raw R&D headcount and physical prototyping budget. If the company successfully scales its AI agent and simulation capabilities, it could bring highly optimized inverters, e-motors, and thermal systems to market faster and cheaper than competitors still relying on traditional, computationally heavy physical modeling. Competitors should monitor the company's academic partnerships and patent filings related to virtual sensors and electro-thermal neural networks.

Priority 2: Standardizing and protecting core embedded software architectures (AUTOSAR)

The company views embedded software—specifically Basic Software (BSW), microcontroller interfacing, and application software—as a non-negotiable core competency that must be owned, developed, and tested internally.

Uploaded-document evidence: The company is investing in machine learning algorithms specifically for propulsion management systems to optimize the performance of connected vehicles.[1] The 2025 Annual Report emphasizes that the company's integrated drive modules (iDMs) contain full software offering functional safety and cybersecurity, a capability built on over 40 years of automotive software experience.[1]

Public-source evidence: The company employs dedicated software technical leaders to manage AUTOSAR architectures, ECU hardware abstraction, and ASPICE compliance for advanced Infineon microcontrollers.[22] Teams utilize model-based design (Simulink/TargetLink) to write and test embedded C code for power electronics [22], while utilizing Siemens Polarion and PTC Windchill to manage the complex product lifecycle and software requirements.[25]

Assessment:

The company is attempting to ensure that its physical hardware (e-motors, clutches, transfer cases) remains indispensable to OEMs by tightly coupling it with highly reliable, ASPICE-certified control software. What is known is that the company has a mature, scaled, and globally distributed team executing model-based design and unit testing. What remains uncertain is how successfully the company can integrate new AI-driven control algorithms (like neural-network BLDC motor control explored in Landskrona) into the rigid, highly deterministic safety requirements of the AUTOSAR framework.

Implication for a competitor: Competitors cannot treat the company merely as a metal-bender or legacy hardware supplier. The company's deep internal ownership of BSW and ECU abstraction means it can offer OEMs "plug-and-play" integrated drive modules that simplify the OEM's own software architecture. Competitors must ensure their own embedded software teams are operating at equivalent ASPICE maturity levels, or risk being relegated to supplying commoditized hardware while the company captures the higher-margin software integration value.

Priority 3: Utilizing off-the-shelf enterprise and manufacturing digital platforms to strip overhead

Outside of core product engineering, the company aggressively prioritizes the use of third-party

SaaS and industrial IoT platforms to drive operational efficiency and reduce structural costs.
Uploaded-document evidence: The 2025 Annual Report details a 2023 Structural Costs Plan aimed at reducing structural costs in Foundational products, expected to yield $80 million to $90 million annually by 2027, alongside a 2024 plan to reduce PowerDrive Systems costs.[1]
Public-source evidence: To achieve operational and financial efficiency, the company relies heavily on vendor integrations. It deployed Supervizor to automate SAP invoice auditing [32], Nicus for IT financial management to fix a broken 10-month budget cycle [4], Cognex for machine vision quality inspection in South Korea [28], and Siemens Insights Hub for global energy tracking across 63 facilities.[31]
Assessment:
The company is trying to aggressively strip out back-office and factory-floor inefficiencies to protect margins in a cyclical automotive market. By purchasing mature solutions, the company avoids the massive capital drain of building bespoke IT infrastructure or proprietary computer vision models. What is known is that these deployments are generating immediate returns (e.g., cash recovery from Supervizor, consolidated budgeting from Nicus). What remains uncertain is whether this highly fragmented vendor landscape (SAP, ServiceNow, Nasuni, Siemens, Cognex, AWS) will create long-term data silos that prevent the company from achieving a unified, interoperable enterprise data architecture.
Implication for a competitor: Competitors should observe that the company is actively automating its administrative and manufacturing overhead through smart vendor procurement. If competitors are still relying on manual SAP auditing, spreadsheet-based IT budgeting, or human visual inspection for complex EV components, they will carry a distinct structural cost disadvantage. However, competitors who can successfully unify their enterprise and manufacturing data into a single proprietary data lake may eventually outmaneuver the company's reliance on disparate third-party vendor platforms.

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